

Section 13.5: Transformers

Mini Investigation: Observing Transformers at Work, page 606

A. Answers may vary. Sample answer:

The transformer did not work with DC because the multimeter connected to the primary coil showed a voltage but the multimeter connected to the secondary coil did not show any voltage.

B. Answers may vary. Sample answer:

The AC voltage on the primary coil was less than the AC voltage on the secondary coil.

C. Answers may vary. Sample answer:

The voltage was increased, so the transformer is a step-up transformer.

Tutorial 1 Practice, page 608

1. **Given:** $V_p = 240 \text{ V}$; $N_p = 550$; $N_s = 110$

Required: V_s

$$\text{Analysis: } \frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$V_s = \frac{V_p N_s}{N_p}$$

Solution:

$$V_s = \frac{V_p N_s}{N_p}$$

$$= \frac{(240 \text{ V})(110)}{550}$$

$$V_s = 48 \text{ V}$$

Statement: The voltage in the secondary coil is 48 V.

2. **Given:** $V_p = 31.0 \text{ V}$; $N_p = 211$; $N_s = 844$

Required: V_s

$$\text{Analysis: } \frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$V_s = \frac{V_p N_s}{N_p}$$

Solution:

$$V_s = \frac{V_p N_s}{N_p}$$

$$= \frac{(31.0 \text{ V})(844)}{211}$$

$$V_s = 124 \text{ V}$$

Statement: The voltage in the secondary coil is 124 V.

Tutorial 2 Practice, page 608

1. **Given:** $V_p = 240 \text{ V}$; $V_s = 12 \text{ V}$; $I_p = 0.15 \text{ A}$

Required: I_s

$$\text{Analysis: } \frac{I_s}{I_p} = \frac{V_p}{V_s}$$

$$I_s = \frac{V_p I_p}{V_s}$$

Solution:

$$I_s = \frac{V_p I_p}{V_s}$$

$$= \frac{(240 \cancel{\text{ V}})(0.15 \text{ A})}{12 \cancel{\text{ V}}}$$

$$I_s = 3.0 \text{ A}$$

Statement: The current in the secondary coil is 3.0 A.

2. **Given:** $V_p = 620 \text{ V}$; $V_s = 12\,000 \text{ V}$; $I_s = 1.3 \text{ A}$

Required: I_p

$$\text{Analysis: } \frac{I_s}{I_p} = \frac{V_p}{V_s}$$

$$I_p = \frac{V_s I_s}{V_p}$$

Solution:

$$I_p = \frac{V_s I_s}{V_p}$$

$$= \frac{(12\,000 \cancel{\text{ V}})(1.3 \text{ A})}{620 \cancel{\text{ V}}}$$

$$I_p = 25 \text{ A}$$

Statement: The current in the primary coil is 25 A.

Section 13.5 Questions, page 609

1. Transformers need an alternating current to operate continuously because they operate according to the law of electromagnetic induction, which states that a changing magnetic field is required to induce a current. The alternating current in the primary coil causes a changing magnetic field to be induced in the core, which then induces an alternating current in the secondary circuit.

2. A step-up transformer has more windings on its secondary circuit than on its primary circuit. A step-down transformer has more windings on its primary circuit than on its secondary circuit.

3. If the voltage and current increase in a step-up transformer, there would be more power coming out of the secondary coil than power going into the primary coil. Since power is equivalent to energy in equal amounts of time this would violate the law of conservation of energy.

4. The voltage would increase since it is directly proportional to the number of windings on the secondary coil. The current would decrease since it is inversely proportional to the number of windings on the secondary coil.

5. A device that has the same number of windings on both the primary coil and the secondary coil would not be capable of raising or lowering AC voltage, so it should not be classified as a transformer.

6. Transformers are not 100 % efficient because some of the energy is transformed into unusable thermal energy in the coils, as well as sound energy.

7. The number of windings on the primary coil of the transformer is 1.5 times greater than on the secondary coil, so $\frac{N_p}{N_s} = 1.5$. Using the equations

related to transformers, solve for V_s and I_s :

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\frac{V_p}{V_s} = 1.5$$

$$1.5 \times V_s = V_p$$

$$\begin{aligned} V_s &= \frac{V_p}{1.5} \\ &= \frac{12.0 \text{ V}}{1.5} \\ V_s &= 8.0 \text{ V} \end{aligned}$$

$$\frac{I_s}{I_p} = \frac{N_p}{N_s}$$

$$\frac{I_s}{I_p} = 1.5$$

$$\begin{aligned} I_s &= 1.5 \times I_p \\ &= 1.5 \times (3.0 \text{ A}) \end{aligned}$$

$$I_s = 4.5 \text{ A}$$

So $V_s = 8.0 \text{ V}$ and $I_s = 4.5 \text{ A}$.

8.

Transformer	V_p	V_s	N_p	N_s	I_p	I_s	Step-up or step-down?
1	12 V	120 V	100	1000	1.2 A	0.12 A	step-up
2	660 V	110 V	600	100	150 mA	900 mA	step-down
3	30 V	5 V	120	20	0.28 A	1.68 A	step-down

9. Substitute the given values for V_p and V_s in the relevant equation related to transformers to find the ratio of the windings:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

$$V_p = 120 \text{ V}; V_s = 15.0 \text{ V}$$

$$\begin{aligned} \frac{N_p}{N_s} &= \frac{V_p}{V_s} \\ &= \frac{120 \cancel{\text{V}}}{15.0 \cancel{\text{V}}} \end{aligned}$$

$$\frac{N_p}{N_s} = 8$$

So the ratio of the primary windings to secondary windings in the transformer is 8 : 1.